Signals and Systems

Formulae

Standard Laplace Transforms

Signal	Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$\delta(t-\tau),\ \tau \geq 0$	$e^{-s\tau}$	$\forall s$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
$\cos(w_1 t)u(t)$	$\frac{s}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$\sin(w_1 t)u(t)$	$\frac{w_1}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}\cos(w_1 t)u(t)$	$\frac{s+a}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$e^{-at}\sin(w_1 t)u(t)$	$\frac{w_1}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$

Standard z-transform pair

$$x[n-n_0] \overset{\mathrm{z}}{\longleftrightarrow} z^{-n_0}X(z) \quad \text{ROC: } R_x \text{ but may exclude } z=0 \text{ or } \infty$$

1. *This question caries 40% of the marks.*

a) Explain the meaning and the key properties of

i) a quantized signal, [2]

ii) a continuous-time signal, [2]

iii) a discrete-time signal. [2]

Solution:
Quantized signals have a finite set of possible amplitudes; continuous-time signals are defined for any and all time; discrete-time signals are defined only at sampling instants.

b) Give an example of a random signal encountered in the physical world. [2]

Solution:
All reasonable suggestions are acceptable, providing that randomness is a clear property. Eg. acoustic noise.

c) For each of the following continuous-time signals, state whether they are causal or noncausal, periodic or nonperiodic, and even, odd or non-symmetric. Give justifications for your statements. If periodic, state the period.

i) $x(t) = \sin^2(t/2 + \pi)$ [2]

Solution:
Non-causal, even symmetric, periodic with period 2π.

ii) $x(t) = \int_{-\infty}^{t} \delta(\tau) d\tau$ [2]
where the function $\delta(t)$ is defined as usual.

Solution:
Causal, non-periodic, non-symmetric.

iii) $x(t) = \begin{cases} 3e^{-0.5t} & t \geq 0 \\ 0 & \text{otherwise.} \end{cases}$ [2]

Solution:
Causal, non-periodic, non-symmetric.

d) Consider the signal

$$x(t) = \begin{cases} 1-t & 0 \leq t \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

For each of the following signals $y(t)$, find an expression for $y(t)$ not involving x. Describe the sequence of steps that must be taken to form $y(t)$ from $x(t)$. Show the sequence of steps in illustrative sketches.

i) $y(t) = x(0.25t + 1)$ [3]

ii) $y(t) = x(-2t + 2)$ [3]

Solution:
(i)

$$y(t) = 1-(0.25t+1) \quad 0 \le (0.25t+1) \le 1$$
$$= \begin{cases} -0.25t & -4 \le t \le 0 \\ 0 & \text{otherwise.} \end{cases}$$

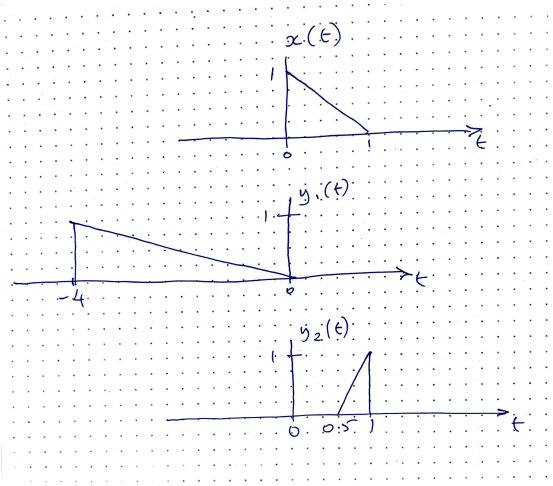
$y(t)$ is formed from $x(t)$ by shifting left by 1 and expanding by a factor of 4.

(ii)

$$y(t) = 1-(-2t+2) \quad 0 \le (-2t+2) \le 1$$
$$= \begin{cases} 2t-1 & 1 \le t \le 0.5 \\ 0 & \text{otherwise.} \end{cases}$$

$y(t)$ is formed from $x(t)$ by shifting left by 2, reflecting the time axis around $t = 0$, and compressing by a factor of 2.

In both cases, the sketches only need to show the labelling of the time and amplitude axes and the shape of the function $y(t)$ positioned correctly on the axes. Rough sketches are fully acceptable, providing these key elements are clear.



[*1 mark for expression, 1 mark for description, 1 mark for sketch*]

e) A real-valued signal $x(t)$ has Fourier transform $X(j\omega)$. A second signal $q(t)$ has Fourier transform $Q(j\omega)$. These signals are related as

$$q(t) = x(-t).$$

i) Derive an expression for $Q(j\omega)$ in terms of $X(j\omega)$, and show your working. [3]

ii) State how the magnitude spectra of $X(j\omega)$ and $Q(j\omega)$ are related. [1]

Solution:

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \\ Q(j\omega) &= \int_{-\infty}^{\infty} x(-t)e^{-j\omega t}dt \\ &= \int_{\infty}^{-\infty} -x(t)e^{j\omega t}dt \\ &= \int_{-\infty}^{\infty} x(t)e^{j\omega t}dt = X^*(j\omega). \end{aligned}$$

The magnitude spectrum is unchanged by the complex conjugation.

f) The discrete-time signal $y[n]$ is generated from a discrete-time signal $x[n]$, so that

$$y[n] = x[n] + ay[n-2]$$

where a is a scalar constant.

Transform this operation into the z-transform domain so as to obtain the corresponding expression for $Y(z)$. Write your expression so that any powers of z are only positive powers of z. [4]

Solution:

$$Y(z) = X(z)\frac{z^2}{z^2 - a}.$$

g) Consider the signal $x_1[n]$ shown in Fig. 1.1. The magnitude and phase spectra are computed using a 16-point DFT and are shown in Fig. 1.2. The sampling frequency used in this question is 8 kHz.

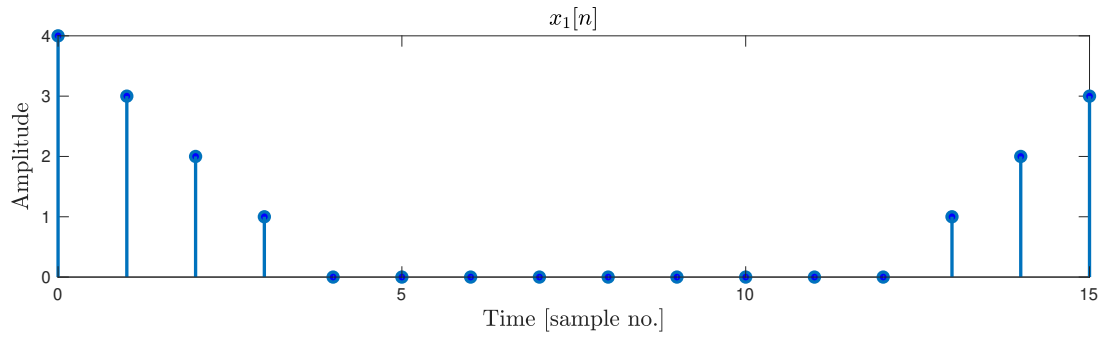


Figure 1.1

i) What is the frequency resolution, Δf, of the 16-point DFT? [1]

Solution: $\Delta f = \frac{f_s}{N} = \frac{8000}{16} = 500$ Hz.

ii) Is the frequency domain signal representation, $X_1[k]$, purely real, purely imaginary, or complex? Justify your answer using properties of the time-domain signal. [2]

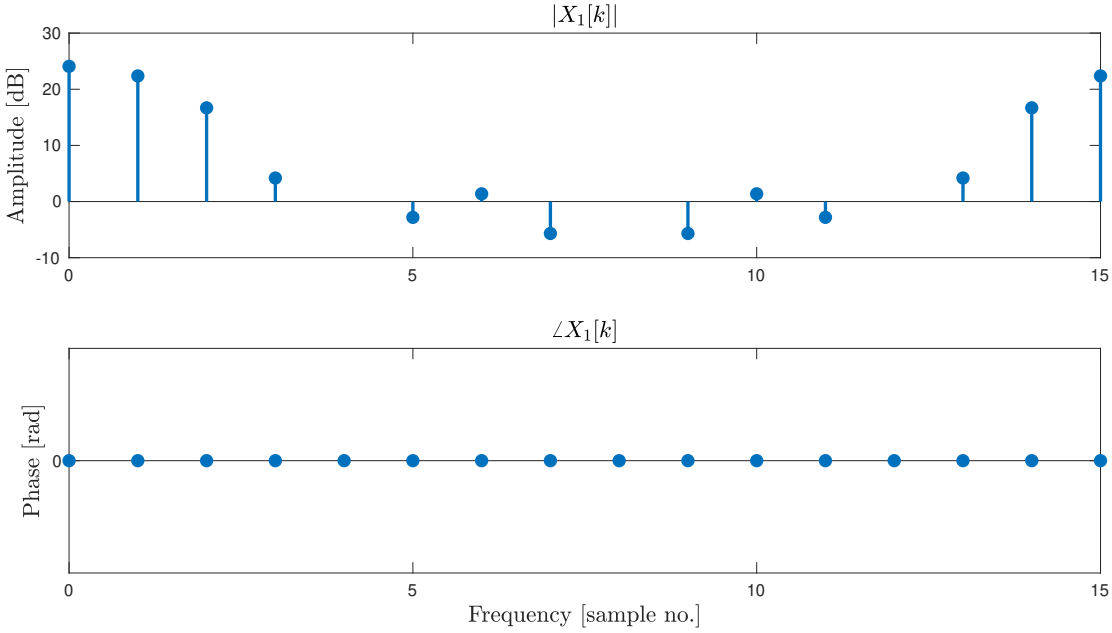


Figure 1.2

Solution: Purely real signal due to the symmetry of the time domain signal (1 mark). The phase is zero since the time-domain signal is even (1 mark).

iii) Now consider the delayed signal $x_2[n]$ plotted in Fig. 1.3. The 16-point DFT of $x_2[n]$ is taken to obtain $X_2[k]$. Describe similarities and differences between the spectra $X_2[k]$ and $X_1[k]$, stating any properties of the Fourier transform supporting your argument.

[6]

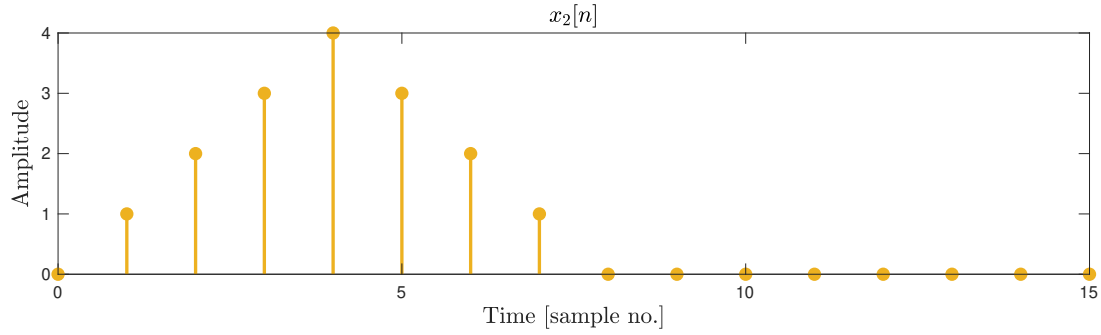
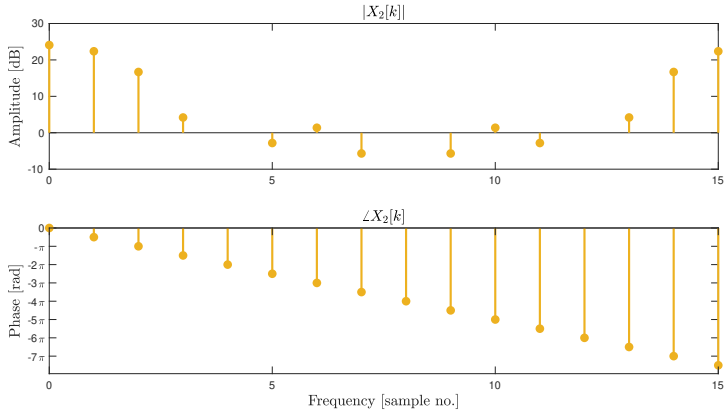


Figure 1.3

Solution: $X_2[k]$ is given by the plot below. The answer should include any 6 marks from

- *Time shifting property of DFT* (1 mark) $x[n-D] \leftrightarrow e^{2\pi kD/N}X[k]$ (1 mark).
- The magnitude spectrum is unchanged by the delay (1 mark).
- The 4-sample delay introduces a linear phase with a negative slope (1 mark). The slope is given by $-2\pi * 4/16 = -\pi/2$ (2 marks).
- $X_1[k]$ and $X_2[k]$ are both $sinc^2$ functions since both $x_1[n]$ and $x_2[n]$ are triangular functions (1 mark). This can be stated as a common transform pair, or derived from the convolution property (1 mark):

$$\text{rect}(t) * \text{rect}(t) = \Delta(t) \leftrightarrow \text{sinc}(\omega/2) \cdot \text{sinc}(\omega/2).$$



iv) Now consider the signal $x_3[n]$ plotted in Fig. 1.4. The 16-point DFT of $x_3[n]$ is taken to obtain $X_3[k]$. Describe the phase of $X_3[k]$ and give a mathematical expression for $\angle X_3[k]$. [3]

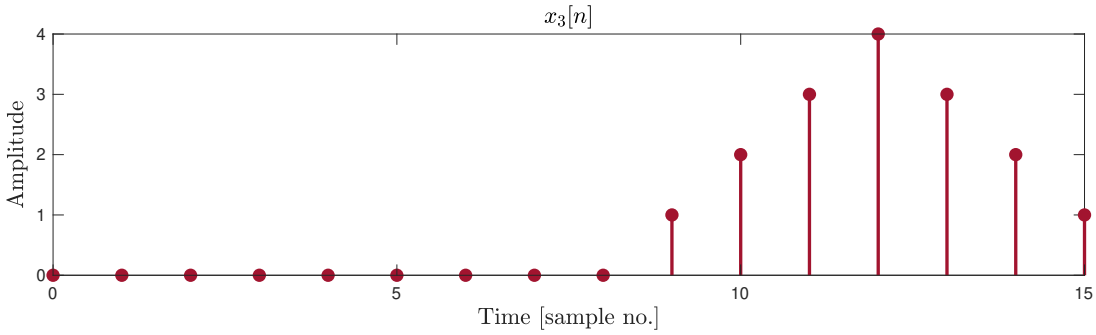


Figure 1.4

Solution: The phase is linear with a positive slope (1 mark) given by $2\pi * 4/16 = \pi/2$ (i.e. $\angle X_3[k] = \pi/2k$) (1 mark). In the DFT, a delay of 12 samples is equivalent to an advance by 4 samples (1 mark). For answers giving slope of $-3\pi/2$ (2 marks). Max 3 marks.

2. *This question caries 30% of the marks.*

a) Consider $y(t)$ to be the product of two continuous-time signals such that

$$y(t) = x(t)p(t).$$

Let $Y(\omega)$, $X(\omega)$ and $P(\omega)$ denote the Fourier transforms of $y(t)$, $x(t)$ and $p(t)$ respectively.

i) State the Multiplication Property of the Fourier transform and hence write down an expression for $Y(\omega)$ in terms of $X(\omega)$ and $P(\omega)$. [4]

ii) Let $p(t) = \cos\omega_0 t$. Write down the corresponding expression for $P(\omega)$. Hence, express $Y(\omega)$ as a function of only $X(\omega)$.[4]

Solution:
To find an expression for $Y(\omega)$, the modulation property can be written

$$y(t) = x(t)p(t) \leftrightarrow Y(\omega) = \frac{1}{2\pi}\left[X(\omega) * P(\omega)\right]$$

where $*$ indicates convolution.

Given $p(t)$, the Fourier transform is

$$P(\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

and

$$Y(\omega) = \frac{1}{2}\big(X(\omega - \omega_0) + X(\omega + \omega_0)\big).$$

b) Consider the signal

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}.$$

i) Deduce the key property of $x(t)$, the meaning of ω_0 and the units of ω_0. [3]

ii) Write a corresponding expression for $x^*(t)$ where the superscript $*$ indicates the complex conjugate. [3]

iii) Show that $x(t)$ is real if $a_k^* = a_{-k}$. [2]

iv) Now consider a particular $x(t)$ with:
$a_{-3} = \frac{1}{4}$, $a_{-2} = \frac{1}{3}$, $a_{-1} = \frac{1}{2}$, $a_0 = 1$, $a_1 = \frac{1}{2}$, $a_2 = \frac{1}{3}$, $a_3 = \frac{1}{4}$,
and $a_k = 0$ for $|k| > 3$.

Write $x(t)$ as a sum of sine and cosine functions and hence draw an indicative sketch of $x(t)$. [6]

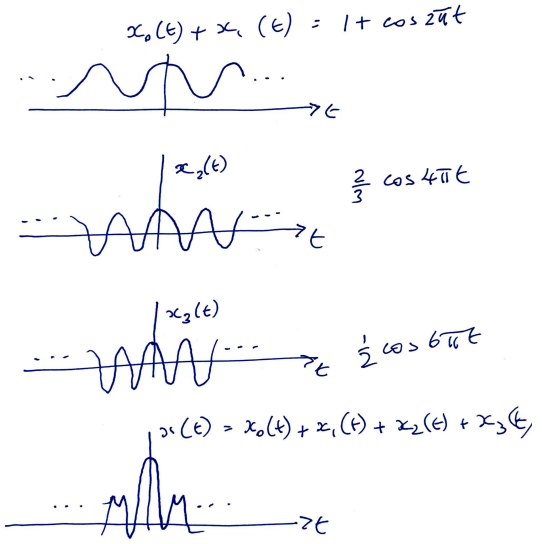
Solution:
(i) periodic, fundamental (angular) frequency in rads/s.
(ii)

$$x^*(t) = \sum_{k=-\infty}^{\infty} a_k^* e^{-jk\omega_0 t}$$

(iii) Replace k with $-k$ in the expression for $x^*(t)$ and compare with the expression for $x(t)$ to see that we require $a_k = a_{-k}^*$ or then $a_k^* = a_{-k}$.
(iv)

$$\begin{aligned} x(t) &= 1 + \frac{1}{2}(e^{j2\pi t} + e^{-j2\pi t}) + \frac{1}{3}(e^{j4\pi t} + e^{-j4\pi t}) + \frac{1}{4}(e^{j6\pi t} + e^{-j6\pi t}) \\ &= 1 + \cos(2\pi t) + \frac{2}{3}\cos(4\pi t) + \frac{1}{2}\cos(6\pi t). \end{aligned}$$

Sketch not required to be accurate but needs to show the summation of the various components. For example:



c) A particular LTI system with input signal $x(t)$ and output signal $y(t)$ is described by

$$\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t).$$

i) Show that the frequency response of this system is

$$H(j\omega) = \frac{j\omega + 2}{(j\omega)^2 + 4j\omega + 3}.$$

Clearly explain each step of your solution. [2]

ii) For an input signal $x(t) = e^{-t}u(t)$, write down its spectrum $X(j\omega)$. [1]

iii) Write down a frequency domain expression for the output spectrum, $Y(j\omega)$, when the input signal is as defined in part ii). [2]

iv) Find the time domain expression for the output signal, $y(t)$, using partial fraction expansion to determine the inverse Fourier transform of $Y(j\omega)$. [3]

Solution:
(i) The solution must mention the differentiation property of the Fourier transform, either explicitly or implicitly, or use the corresponding property of the Laplace transform with the assumption of zero imaginary part of s.
(ii)

$$X(j\omega) = \frac{1}{1+j\omega}$$

(iii)

$$Y(j\omega) = H(j\omega)X(j\omega) = \frac{j\omega+2}{(j\omega+1)^2(j\omega+3)}.$$

(iv)

$$y(t) = \frac{1}{4}e^{-t}u(t) + \frac{1}{2}te^{-t}u(t) - \frac{1}{4}e^{-3t}u(t).$$

3. *This question caries 30% of the marks.*

a) The discrete Fourier transform for a signal $x[n]$ can be written

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} \quad k = 0, 1, \ldots, N-1.$$

i) State the meaning of the variables n, N, and k. [3]

ii) Explain the meaning of $X(3)$ for the case $N = 8$. Is $X(3)$ a real-valued or complex-valued quantity? [3]

iii) Given that $x[n] = [1, -1, 1, -1, 1, -1, 1, -1]$, determine $X(3)$ and $X(4)$. [4]

Solution:
Time index n, number of time domain samples N, frequency index k.
$X(3)$ is the complex-valued (1 mark) frequency domain representation of $x[n]$ at 3/8 of the sampling frequency (2 marks).

For $k = 3$, $\exp(-j2\pi kn/N) =$
$1 + j0$
$-0.7071 - j0.7071$
$0 + j$
$0.7071 - j0.7071$
$-1 + j0$
$0.7071 + j0.7071$
$0 - j$
$-0.7071 + j0.7071$
so that the sum of the product with $x[n]$ = 0.

For $k = 4$, $\exp(-j2\pi kn/N) =$
$1 + j0$
$-1 + j0$
$1 + j0$
$-1 + j0$
$1 + j0$
$-1 + j0$
$1 + j0$
$-1 + j0$
so that the sum of the product with $x[n]$ = 8.

b) i) Consider the causal discrete-time system illustrated in Fig. 3.1.

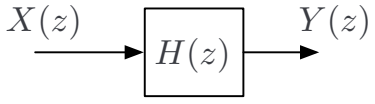


Figure 3.1

Given that

$$H(z) = \frac{z}{z^2 - (0.1 - a)z - 0.1a},$$

find the range of values of a for which the system is stable.

[4]

Solution:
Real roots at $z = 0.1$ and $z = -a$. For stability, all roots must be within the unit circle on the z-plane, hence the range is given by $|a| < 1$.

ii) Using the standard z-transform pair given in the formulae list above, or otherwise, find the difference equation for $y[n]$ for the system of part i).

[4]

Solution:

$$Y(z)[1-(0.1-a)z^{-1}-0.1az^{-2}] = X(z)z^{-1}$$
$$y[n] = x[n-1]+(0.1-a)y[n-1]+0.1ay[n-2]$$

iii) Now consider the causal discrete-time system illustrated in Fig. 3.2.

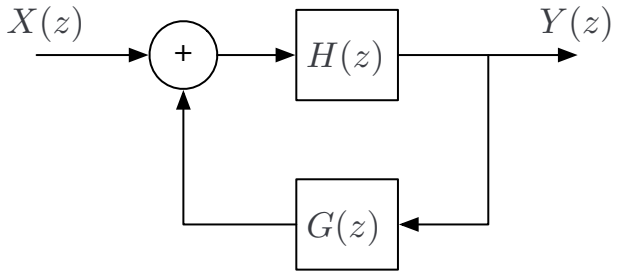


Figure 3.2

Given that

$$H(z) = \frac{z}{z-0.1} \quad \text{and} \quad G(z) = kz^{-1},$$

find the range of values of k for which the system is stable.

[4]

Solution:

$$Y(z) = \left[X(z)+kz^{-1}Y(z)\right]\frac{z}{z-0.1}$$
$$\frac{Y(z)}{X(z)} = \frac{z}{z-0.1}\Big/\left[1-\frac{k}{z-0.1}\right] = \frac{z}{z-0.1-k}.$$

For stability, we require the real pole to be inside the unit circle so that

$$-0.9 < k < 1.1$$

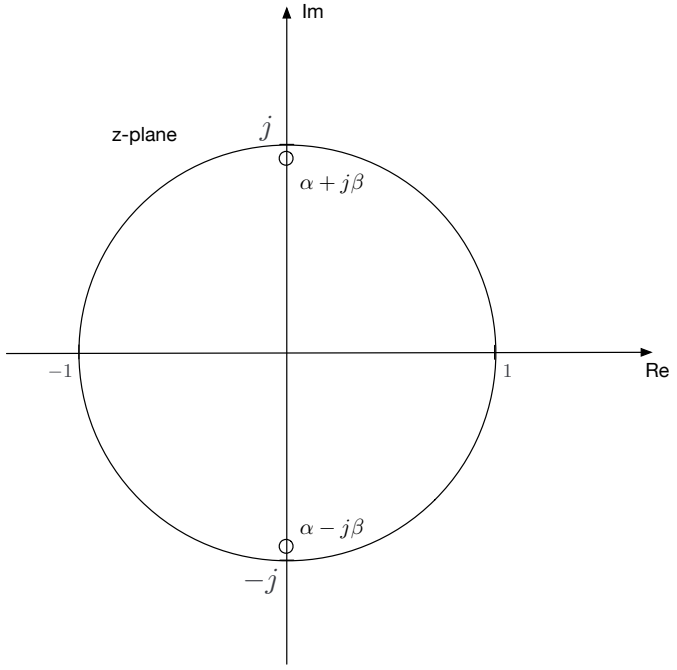
c) A causal, discrete-time, 2nd-order FIR filter $P(z)$ operates with a sampling frequency of 8 kHz. The filter is designed to attenuate an interfering

signal with a frequency of 2 kHz.

i) Deduce and sketch a z-plane plot of a suitable filter's system function $P(z)$. [4]

ii) Determine and write down the corresponding z-domain expression for $P(z)$ for the filter you deduced in part i). [4]

Solution:



For a general second order FIR,
$P(z) = 1 + b_1 z^{-1} + b_2 z^{-2}$ (2 marks)
For zeros at $\alpha \pm j\beta$:
$P(z) = [z - (\alpha + j\beta)][z - (\alpha - j\beta)]$ (1 mark).
For roots at $\Omega = 0 \pm \pi/2$:
$P(z) = z^2 + \beta^2$ (1 mark)